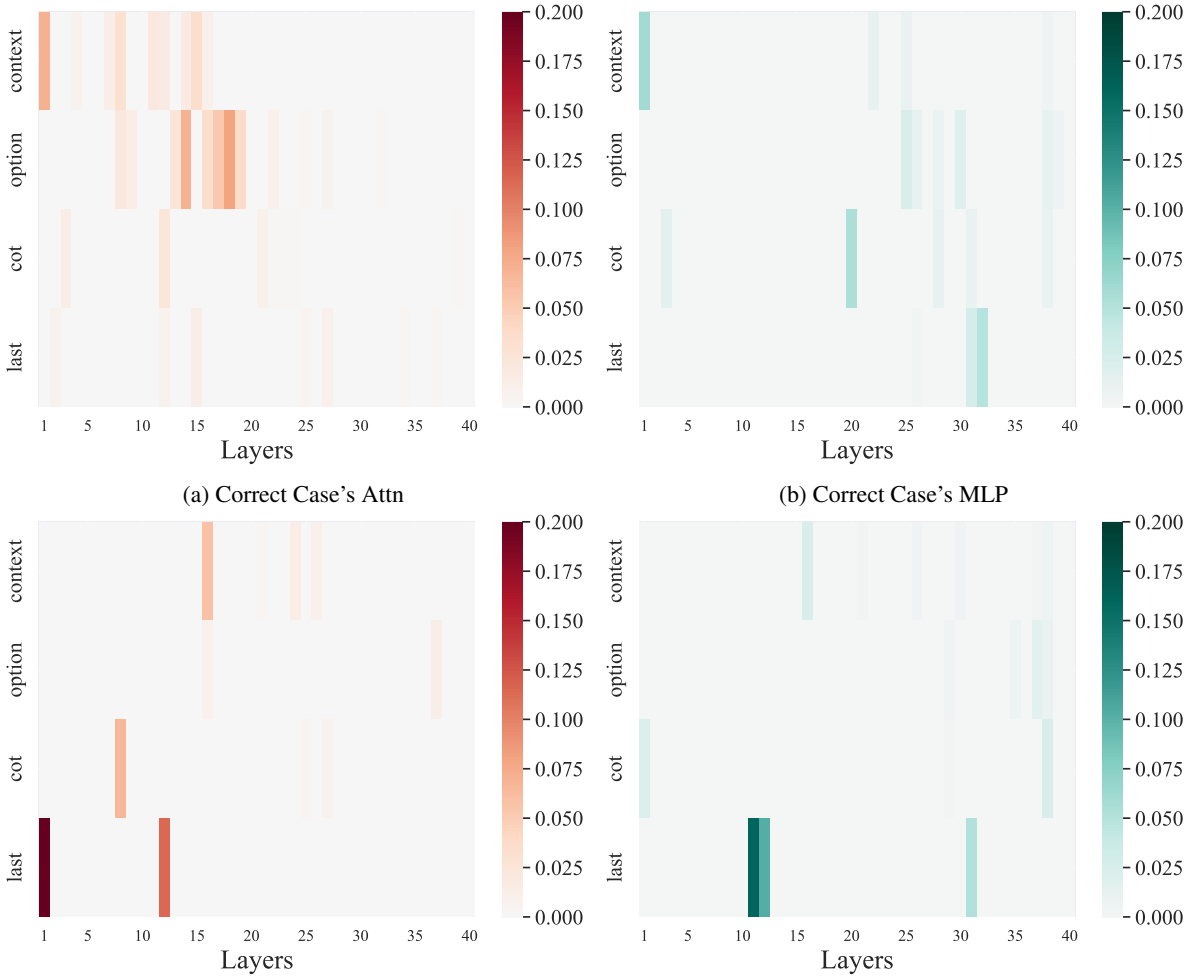


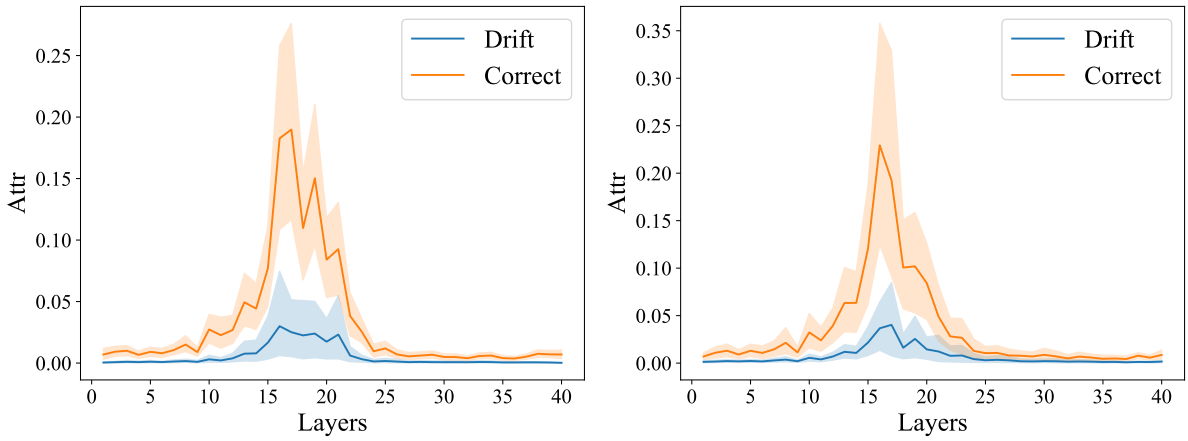
(a) Correct Case's Attn

(b) Correct Case's MLP

(c) Drifting Case's Attn

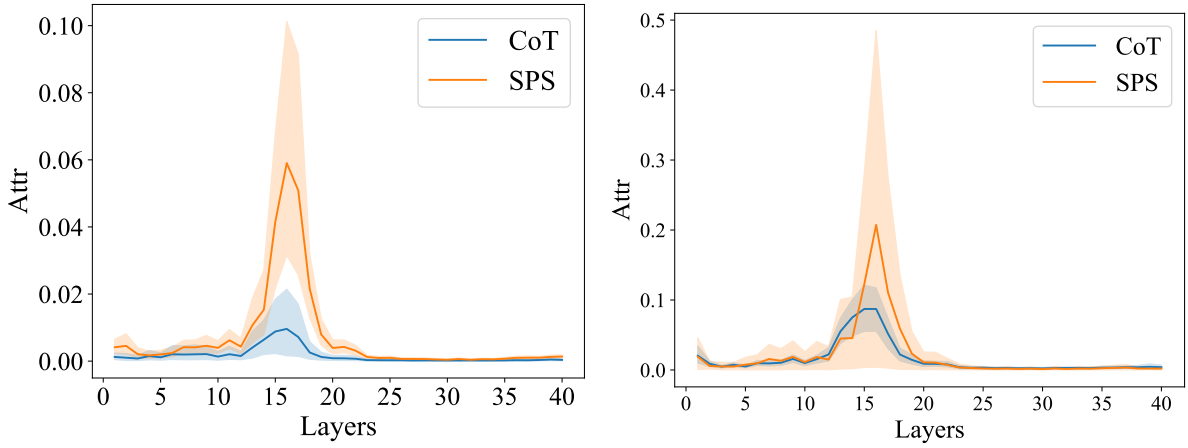
(d) Drifting Case's MLP

Figure 14: Intervention tracing results on CSQA in correct and drifting answering cases (Baichuan2-13B).



(a) Winogrande (b) CSQA

Figure 15: Attribution tracing results on Baichuan2-13B during the answer generation stage.



(a) Winogrande (b) CSQA

Figure 16: Information flow comparison on CSQA after applying our two methods.

System Instruction: You are a helpful, respectful and honest assistant. You should use your reasoning abilities to given the reason for the answer to the given questions. Your response should be in this form: 'Reason: {reason}'.

Example Input: Question: {QUESTION}\nAnswer: {ANSWER}
Example Output: Reason: {COT}

Input: Question: {QUESTION}\nAnswer: {ANSWER}
Output:

Figure 17: Prompts for correct CoT generation.

System Instruction: You are a helpful, respectful and honest assistant. You should use your reasoning abilities to answer the given questions in reasoning tasks. You should reply the correct rationales and the answer. Your response should be in this form: '{reason} So the answer is: ({option}) {answer}'. If you don't know the answer to a question, please reply 'Answer: None'.

Question: The test was hard for Samuel but a breeze for Randy , since _ had failed to study for it.\n(1) Samuel (2) Randy
Answer: To pass a test, a person need to study for it. If a person feel the test like a breeze, it means the test is easy for him. A person feels the test easy, because he studies hard for it. Since we know that Samuel feel the test very hard, she may fail to study for it. So the answer is: (1) Samuel.

Question: Kyle slowly wormed their way into the life of Derrick, because _ was good and manipulating people.\n(1) Kyle (2) Derrick
Answer: A person wormes his way into other's life, because he is friendly and approachable. A friendly person is considered good. A person is seen as manipulating people, that means he like to interact with others and others like him. Since Kyle slowly wormed their way into the life of Derric, Kyle will be seen as good and manipulating people. So the answer is: (1) Kyle.

Question: Donald was very grounded but Michael often got lost in their daydreams. _ was very capricious all the time.\n(1) Donald (2) Michael
Answer: A person is grounded means he works hard and does not like to fantasize. A person often gets lost in his daydreams, he is seen as unrealistic and egocentric. A person is capricious all the time means he does everything only according to his own ideas. Since Michael often gets lost in their daydreams but Donald does not, Michael is seen as very capricious all the time. So the answer is: (2) Michael.

Question: After stopping when running, Betty was able to help Rachel treat their asthma because _ has it too.\n(1) Betty (2) Rachel
Answer: A person who has a mild disease may knows how to treat it. Since Since Betty was able to help Rachel treat their asthma, Betty may has it, too. So the answer is: (1) Betty.

Question: Jeffrey was sick with the stomach flu and not hungry, but Christopher was starving. _ ordered food.\n(1) Jeffrey (2) Christopher
Answer: Stomach flu is a viral infection in the digestive system. A person is starve so that he need to eat something next. A person is not hungry means he does not want to eat something. Ordering food means to buy food online. A person may eat food after ordering it. Since Christopher was starving and Jeffery was sick and not hungry, Christopher tended to order food. So the answer is: (2) Christopher.

Figure 18: 5-shot prompts for Winogrande.